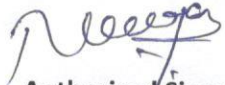




Plot No. 4300-4301 G.I.D.C Phase-III Dared, Jamnagar-361004 (Gujarat) Ph:0288-2730060
e-mail. patelmetalalloys@gmail.com GST No.24AAVFP6629H1Z9
Manufacture All Kind of Brass Rods, Section, Hollow Rods.

TEST CERTIFICATE

Customer:		MTC No.	3216	Date:	16-12-2023
METAL MECH ENGINEERING		Invoice No.	PMA/1411	Date:	16-12-2023
JAMNAGAR		P.O. No.	2EX/0460/23/24	Date:	
		Qty. (Kgs.).			
We hereby certify that above material conforms to the following Chemical composition & Physical properties.					
PARTICULARS	As Required by Specification	Specification of Supplied Material			
I. CHEMICAL COMPOSITION	EN 12164 CW614N CUZN39PB3	Size	Size	Size	Size
		60 X 42 MM ROUND HOLLOW ✓	51 X 36 MM ROUND HOLLOW ✓		
		Heat No.	Heat No.	Heat No.	Heat No.
		C-251 ✓	C-254 ✓		
		304.150 Kgs.	273.700 Kgs.		
1. COPPER (Cu) %	57.00 – 59.00	57.70	57.68		
2. LEAD (Pb) %	2.50-3.50	2.80	2.84		
3. IRON (Fe) %	0.30 Max	0.25	0.25		
4. ZINC (Zn) %	Remainder	Remaining	Remaining		
5. ALUMINIUM (Al) %	0.05 Max	0.02	0.04		
6. MANGANESE (Mn) %	-	-	-		
7. NICKLE (Ni) %	0.30 Max	0.28	0.22		
8. TIN (Sn) %	0.30 Max	0.27	0.29		
9. SILICON (Si) %	-	-	-		
10. PHOSPHORUS (P) %	-	-	-		
11. SULPHUR (S) %	-	-	-		
12. TOTAL IMPURITIES %	0.20 Max	0.07	0.07		
II. PHYSICAL PROPERTIES					
1. UTS – Min (Mpa)	430 MIN	439	439		
2. ELONGATION %	20 MIN	26	26		
3. HARDNESS (HV5)	-	-	-		
4. IZOD	-	-	-		
5. 0.2% PROOF STRESS	350 MAX-	342	342		
III. Mercurous Nitrate Test					
6. CONDUCTIVITY	-	-	-		
For. Patel Metal Alloys					
 Authorized Signatory					

